YIJUN LI

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ACADEMIC EMPLOYMENT

Dana-Farber Cancer Institute, Boston, MA *Sept. 2024–present*
Research Fellow, Department of Data Science.
Mentor: Rafael Irizarry.

EDUCATION

University of Michigan, Ann Arbor, MI *Sept. 2024*
Ph.D. in Biostatistics.
Dissertation: *Computational Analysis and Statistical Methods in Spatial Transcriptomics and Neuroimaging*.
Advisors: Jian Kang and Lana Garmire.

University of Michigan, Ann Arbor, MI *May 2019*
M.S. in Biostatistics

Duke University, Durham, NC *May 2017*
B.S. in Statistical Science.
B.S. in Mathematics.

RESEARCH INTERESTS

Spatial transcriptomics; spatial proteomics; single-cell genomics; computational pathology; multimodal data integration; biological networks; tissue architecture; Bayesian modeling; statistical learning; AI.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed journal articles

[1] **Yijun Li**, Ki Sueng Choi, Boadie W. Dunlop, W. Edward Craighead, Helen S. Mayberg, Lana Garmire, Ying Guo, Jian Kang. BSNMani: Bayesian Scalar-on-Network Regression with Manifold Learning. *The Annals of Applied Statistics*. 2026; 20(2):1496-1515. **Young Statistician Award**

[2] Tong Liu, Euiseong Ko, Yiwen Yang, Haowen Wu, Thatchayut Unjitwattana, Suyuan Wang, Jian Kang, **Yijun Li**$^{+}$, Lana X Garmire^{+}. Population-Scale Integration of Spatial Omics Networks for Clinical Prediction and Biological Discovery by SPIN. *Accepted by Cell Genomics*. 2026.

[3] **Yijun Li**, Stefan Stanojevic, Bing He, Zheng Jing, Qianhui Huang, Jian Kang, and Lana Garmire. Adding Highly Variable Genes to Spatially Variable Genes Can Improve Cell Type Clustering Performance in Spatial Transcriptomics Data. *Bioinformatics Advances*. 2026; 6(1):vbaf285

[4] Bing He, Yao Xiao, Haodong Liang, Qianhui Huang, Yuheng Du, **Yijun Li**, David Garmire, Duxin Sun, Lana Garmire. ASGARD is A Single-cell Guided Pipeline to Aid Repurposing of Drugs. *Nature Communications*. 2023 Feb 22;14(1):993.

[5] Lana X. Garmire, **Yijun Li***, Qianhui Huang*, Chuan Xu, Sarah A. Teichmann, Naftali Kaminski, Matteo Pellegrini, Quan Nguyen & Andrew E. Teschendorff. Challenges and perspectives in computational deconvolution of genomics data. *Nature Methods*. 2024 Mar;21(3):391-400.

[6] Stefan Stanojevic, **Yijun Li**, Aleksandar Ristivojevic, and Lana X. Garmire. Computational Methods for Single-Cell Multi-Omics Integration and Alignment. *Genomics Proteomics & Bioinformatics*. 2022 Oct;20(5):836-849.

[7] **Yijun Li**, Stefan Stanojevic, and Lana X. Garmire. Emerging Artificial Intelligence Applications in Spatial Transcriptomics Analysis. *Computational and Structural Biotechnology Journal* 2022;20:2895-2908.

* Equal contribution. $^{+}$ Co-corresponding author.

Preprints

[8] **Yijun Li**, Abdul Rahman Diab, Abdallah Lamane, James R. Lindsay, Scott J. Rodig, Rafael A. Irizarry, William Lotter. Pan-cancer estimation of prognostic immune biomarkers from histopathology using artificial intelligence. *bioRxiv*. 2026.

Manuscripts in preparation

[9] **Yijun Li**, Jeffrey Moffitt, Rafael A. Irizarry. CTMap: a probabilistic framework for defining anatomical regions from spatial cell-type organization. 2026.

[10] **Yijun Li**, Jian Kang. BSNMani+: Scalable Bayesian Scalar-on-Network Regression Model via Manifold Learning. 2026.

TALKS

Partitioning Anatomical Regions in Single-Cell Spatial Omics
Dana-Farber Cancer Institute Data Science Fellowship Showcase for Philanthropy Visit. *October 2025*
Boston, MA.

BSNMani-ST: A Bayesian Model for Linking Spatial Transcriptomics Features to Patient Phenotypes at the Population Scale
JSM 2025. Nashville, TN. *August 2025*
Co-corresponding author; presented by collaborator on behalf of Yijun Li.

Using BSNMani to Link Single Cell Omics Features with Patient Outcome
WNAR 2025. Whistler, BC, Canada. *June 2025*
Co-corresponding author; presented by collaborator on behalf of Yijun Li.

BSNMani+: Scalable Bayesian Scalar-on-Network Regression Model via Manifold Learning
ENAR 2025. New Orleans, LA. *March 2025*

Bayesian Scalar-on-Network Regression with Manifold Learning
International Biometric Conference. Atlanta, GA. *December 2024*

Evaluating Data Integration Tools on Clustering Spatial Transcriptomics Data
Short talk and poster at ACM-BCB. Virtual. *July 2021*
Short talk and poster at ISCB/ECCB. Virtual. *July 2021*
Short talk and poster at MSSISS. Ann Arbor, MI. *February 2021*

POSTERS

CTMap: a quantitative framework for multi-scale spatial tissue architecture
STATGEN. Atlanta, GA. *May 2026*

Bayesian Scalar-on-Network Regression with Manifold Learning
Statistical Methods in Imaging Conference 2024. Indianapolis, IN. *May 2024*
ENAR 2024. Baltimore, MD. *March 2024*
ICSA Applied Statistics Symposium. Ann Arbor, MI. *June 2023*
Modern Statistical and Machine Learning Methods for Big Data. Ann Arbor, MI. *October 2022*
Biostatistics 70th Celebration. Ann Arbor, MI. *May 2022*

Bayesian low rank matrix learning with application to image analysis
BFF5 conference. Ann Arbor, MI. *May 2018*

GRANTS AND AWARDS

NIH K99/R00 Pathway to Independence Award application *October 2026*
National Human Genome Research Institute (NHGRI). Principal Investigator.
Multiscale Tissue Architectural Phenotypes and Their Variation Across Samples for Functional Interpretation of Spatial Genomic Data.
Application no.: 1K99HG016054-01. **Submitted; pending review.**

Young Statistician Award, International Biometric Conference. *December 2024*
Bayesian Scalar-on-Network Regression via Manifold Learning.

Rackham Student Travel Award, ENAR. *March 2024*
Bayesian Scalar-on-Network Regression via Manifold Learning.

Best Poster Award, Modern Statistical and Machine Learning Methods for Big Data. *October 2022*
Bayesian Scalar-on-Network Regression via Manifold Learning.

Best Speed Oral Presentation, Michigan Student Symposium for Interdisciplinary Statistical Sciences (MSSISS). *February 2021*
Evaluating Data Integration Tools on Clustering Spatial Transcriptomics Data.

Teaching and Mentoring

Research Mentor, SPIN Project *2025–2026*

Provided hands-on implementation guidance through remote mentoring meetings three times per week.

- **Suyuan Wang**, master's student in Biostatistics, University of Michigan; subsequently entered the PhD program in Biostatistics at the Medical College of Wisconsin.
- **Tong Liu**, master's student in Biostatistics, University of Michigan.
- **Haowen Wu**, master's student in Biostatistics, University of Michigan.

Rackham Professional Development Diversity, Equity, and Inclusion Certificate *2024*

University of Michigan

Co-instructor, BIOS605 Introduction to SAS Statistical Programming *Fall 2019*

University of Michigan

- Taught SAS programming and data manipulation to first-year master's students.
- Graded homework and final projects.

Graduate Student Instructor

BIOS620 Introduction to Health Data Science *Winter 2021 & 2024*

University of Michigan

- Guided student-designed group projects applying federated learning and distributed computing methods to personal device screen-time data collected throughout the semester.
- Held weekly office hours and graded homework and examinations.

BIOS626 Machine Learning for Health Sciences *Winter 2021*

University of Michigan

- Held weekly office hours and graded homework and examinations.

Professional Service

Journal Reviewer

Nature Communications; Genomics, Proteomics & Bioinformatics; Briefings in Bioinformatics; BMC Bioinformatics; BMC Pharmacology and Toxicology; Discover Applied Sciences; Discover Oncology.

Organizer, Spatial Working Group (SWG) *2026–present*

Department of Data Science, Dana-Farber Cancer Institute.

- Organize monthly meetings for researchers in Dana-Farber and Harvard who are working on or interested in spatial omics to exchange methods, discuss emerging technologies, and share ongoing research.
- Recruit speakers and develop meeting agendas featuring research presentations and methodological discussions.
- Facilitate discussions of relevant papers, computational tools, and analysis challenges, providing opportunities for feedback and collaborative troubleshooting.

Session Chair, ENAR *2025*

Topic: Machine Learning and Causal Inference for Clinical and Genomic Applications.

MSSISS Organizing Committee *2020–2021*

University of Michigan, Ann Arbor, MI.

- Organizing committee member for the 2021 MSSISS conference.
- Invited and corresponded with keynote speaker, Dr. Xihong Lin, and junior keynote speaker, Dr. Raed Al Kontar for the 2020 conference.
- Assisted with budgeting and finding a virtual platform for the 2020 virtual conference in light of COVID-19.